

Zero-Shot Performance of Compact Whisper Variants on Urdu-English Code-Mixed Speech: An Empirical Investigation Using WER and Switch Point Analysis

Muhammad Abdullah Haroon
Department of Computer Science
National University of Computer and Emerging Sciences (FAST-NUCES)
Lahore, Pakistan
abdullaharoon.work@gmail.com

Abstract

Modern speech-to-text systems deliver impressive results for well-resourced languages, yet they falter when confronted with code-mixed utterances prevalent in multilingual societies. This work presents an empirical investigation into the zero-shot transcription capabilities of three compact OpenAI Whisper variants, namely Tiny (39M parameters), Base (74M), and Small (244M), on a purpose-built corpus of 327 Urdu-English code-mixed recordings from 11 speakers. Utterances are stratified into three tiers of mixing intensity (light, moderate, and heavy). We assess recognition quality through Word Error Rate (WER) alongside a diagnostic metric termed Switch Point Error Rate (SPER) that isolates accuracy specifically at language transition boundaries. Three inference configurations are tested: Urdu-forced, English-forced, and automatic language detection. Our experiments demonstrate that: (1) recognition degrades progressively as mixing intensity rises, with up to 36.6% WER inflation between the lightest and heaviest tiers; (2) transcription failures cluster disproportionately at language transition boundaries, with SPER surpassing 0.93 universally; (3) the Small variant paired with English-forced inference attains the lowest error (WER = 0.899); and (4) the choice of language configuration impacts accuracy by as much as 82.6%, though the optimal setting is contingent on model capacity. These findings offer concrete guidance for practitioners deploying speech recognition in code-mixed South Asian settings.

Index Terms—Speech Recognition, Whisper, Code-Mixing, Urdu-English, Word Error Rate, Switch Point Error Rate, Under-Resourced Languages, Zero-Shot Inference, Bilingual ASR

I. Introduction

The landscape of automatic speech recognition (ASR) has been transformed by the advent of transformer-based architectures [14] and large-scale weakly supervised training paradigms [1]. State-of-the-art systems now achieve near-human transcription accuracy for high-resource languages such as English, Mandarin, Spanish, and French, where abundant labeled corpora facilitate robust model training [2]. However, a significant performance gap re-

mains for languages that are underrepresented in training datasets.

Urdu (اردو) represents a striking instance of this imbalance. Despite being spoken by more than 230 million people across Pakistan, India, and global diaspora communities, making it one of the world’s ten largest languages by speaker count [3], Urdu remains severely underserved in both academic ASR research and commercial speech products. This underrepresentation stems from multiple converging factors:

- Scarcity of labeled and transcribed speech datasets for model training
- Complex phonological inventory drawing from Arabic, Persian, and Indic traditions
- Lack of standardized benchmarks and shared evaluation frameworks
- Wide dialectal diversity across Pakistani, Indian, and Deccani variants

The challenge intensifies further due to the widespread phenomenon of **code-mixing** (کوڈمکسنگ), wherein speakers fluidly alternate between two languages within a single utterance or conversational turn. In urban Pakistani settings, educated speakers routinely weave English words, phrases, and clauses into Urdu conversation [4], a behavior observed across workplaces, academic institutions, social media platforms, and broadcast media. Consider the following typical mixed utterance:

“*Kal sham tak saray documents media par nashar ho jayein ge, team ko update kar dein*”

(کل شام تک سارے ڈاکومنٹس میڈیا پر نشر ہو جائیں گے، ٹیم کو اپڈیٹ کر دیں)

Such bilingual input poses cascading challenges for speech recognizers trained on monolingual corpora: sudden phonological transitions at language switch points, lexical confusion where borrowed terms overlap with native vocabulary, and inconsistent prosodic patterns across language segments.

The Whisper model family from OpenAI [1], trained on an extensive 680,000-hour corpus of multilingual web audio, presents a compelling candidate for cross-lingual transcription tasks. While the larger Whisper configurations (Large-V2, Medium) have been extensively studied, the lightweight variants (Tiny at 39M, Base at 74M, and Small at 244M parameters) remain largely unexplored for code-mixed, low-resource language settings. These compact models hold particular practical importance as they are:

- Executable on consumer-grade laptops and mobile devices
- Capable of meeting real-time latency requirements
- Accessible within the limited computing budgets of developing-world institutions
- Scalable for high-volume batch processing without dedicated GPU infrastructure

یہ تحقیقی مقالہ اس بنیادی سوال کا جواب تلاش کرتا ہے کہ محدود حسابی وسائل کے دائرے میں رہتے ہوئے اردو-انگریزی مخلوط تقریر کی خودکار نقل نویسی کس حد تک قابل اعتماد ہو سکتی ہے۔

This study is organized around four core research questions:

- 1) **RQ1:** Does transcription error escalate as code-mixing density increases from light (Category A) through heavy (Category C)?
- 2) **RQ2:** Are recognition failures concentrated at language transition boundaries, or distributed uniformly throughout the utterance?
- 3) **RQ3:** Among the compact Whisper variants, which achieves the strongest performance on code-mixed speech, and what architectural factors explain the gap?
- 4) **RQ4:** Does explicitly specifying the inference language (Urdu vs. English vs. automatic detection) alter recognition accuracy on bilingual input?

The principal contributions of this work are: (1) a purpose-built corpus of 327 Urdu-English code-mixed speech samples with word-level language annotations from 11 speakers; (2) the first rigorous zero-shot evaluation of compact Whisper variants on this bilingual data; (3) the formalization of Switch Point Error Rate (SPER) as a diagnostic metric for quantifying code-mixing robustness; and (4) actionable deployment recommendations for practitioners building speech systems for multilingual South Asian contexts.

II. Literature Review

A. Large-Scale Multilingual Speech Recognition

The emergence of foundation models for speech has reshaped the multilingual recognition landscape. Radford et al. [1] released Whisper, trained on 680,000 hours of weakly labeled web audio spanning 97 languages. The system demonstrated strong cross-lingual generalization

without explicit adaptation, though accuracy varied substantially; well-resourced languages achieved single-digit WER while under-resourced ones often exceeded 50%.

Zhang et al. [2] introduced the Universal Speech Model (USM) at 2 billion parameters, pushing coverage beyond 100 languages. Despite impressive headline numbers, such massive architectures are impractical where GPU resources are scarce. Peng et al. [8] proposed open-source Whisper-style alternatives based on E-Branchformer, observing analogous struggles with under-represented languages.

B. Speaker Diversity and Recognition Robustness

Garnerin et al. [5] studied how speaker background (native versus non-native, regional accent) influences recognition quality even within a single language. North American English speakers consistently obtained lower error rates than British or Australian counterparts. These within-language disparities suggest that cross-lingual transitions during code-mixing would amplify recognition difficulty further.

C. Addressing Data Scarcity for Under-Resourced Languages

Creative data acquisition has received growing interest. Yerbolat and Karakus [6] demonstrated semi-automated corpus construction from audiobooks for Kazakh, partially alleviating the labeled-data bottleneck. Pratap et al. [10] achieved coverage of over 1,000 languages through the Massively Multilingual Speech initiative, though code-mixed scenarios were not considered.

خودکار تقریر شناخت کے میدان میں کم وسائل والی زبانوں کے لیے تحقیق بڑھ رہی ہے، مگر اردو-انگریزی کوڈکسنگ کا شعبہ ابھی تک نظر انداز ہے۔

D. Urdu Speech Recognition Progress

Sharif et al. [3] surveyed Urdu ASR development from classical statistical frameworks (HMMs, GMMs) to contemporary neural approaches. They noted that evaluations of large models such as Whisper-Large and MMS-1B on Urdu exist, but these architectures demand enterprise-grade GPU infrastructure beyond the reach of Pakistani educational institutions and public agencies.

E. Code-Mixing Challenges in Speech Systems

Code-mixing in speech systems presents unique challenges that span acoustic modeling, language modeling, and evaluation methodology [13]. Li et al. [7] investigated asymmetric language modeling strategies for code-switching recognition, while Ali et al. [4] documented Urdu-English switching patterns, observing that transitions tend to occur at predictable syntactic junctions such as noun phrases, technical terminology, and discourse markers. Conneau et al. [9] established FLEURS for cross-lingual speech evaluation, though explicit code-mixing scenarios were absent from its design.

F. Identified Gaps

The literature review surfaces several unaddressed questions that motivate this study:

- 1) How do compact Whisper variants perform on Urdu without any adaptation?
- 2) What is the quantitative relationship between mixing density and recognition accuracy?
- 3) Are transcription failures uniformly distributed or concentrated at specific linguistic locations?
- 4) How does the choice of inference language interact with model capacity for bilingual input?

III. Experimental Design

A. Overview

We adopt a controlled experimental methodology with a 3×3 factorial structure: three model sizes crossed with three language inference settings, producing nine distinct conditions. No fine-tuning or domain adaptation is applied; all results reflect pure zero-shot capability.

B. Corpus Development

1) Participant Selection

Eleven Urdu-English bilingual individuals participated as speakers. All are native Urdu speakers from urban Pakistani backgrounds who naturally incorporate English into daily speech. Participants are categorized into two age brackets (Teenager and Adult) to capture generational variation in code-mixing patterns.

Table I: Participant Information and Recording Counts

Participant	Age Bracket	Recordings
Abdullah Haroon	Adult	30
Ayyan Haroon	Teenager	30
Azib Naem	Adult	30
Faiq Amir	Teenager	30
Ali Hadi	Adult	30
Hassaan Ahmed Khan	Adult	30
M. Abdullah Rafiq	Adult	29 *
M. Moeen Ud Din Ali	Adult	30
Muhammad Musab	Adult	30
Nouman Tanveer	Teenager	30
Zayan Amjad	Adult	28
Total		327

*1 recording excluded due to file corruption

2) Mixing Intensity Tiers

Thirty reference sentences were authored and organized into three tiers (10 per tier) representing progressively denser code-mixing:

Tier A (Mild Mixing, ہلکی آمیزش):

Predominantly Urdu content with scattered English borrowings at noun positions (roughly 15–25% English tokens). Transitions are infrequent and occur at technical terms or proper nouns.

Example: “Kal sham tak saray documents media par nashar ho jayein ge, team ko update kar dein”

کل شام تک سارے ڈاکو منٹس میڈیا پر نشر ہو جائیں گے، ٹیم کو اپڈیٹ کر دیں

Tier B (Moderate Mixing, درمیانی آمیزش):

Roughly balanced distribution with frequent alternation at phrasal boundaries and verb positions (35–50% English tokens).

Example: “Yeh meeting abhi cancel nahi hogi, boss ne personally confirm kiya hai ke agenda finalize ho chuka hai”

یہ میٹنگ ابھی کینسل نہیں ہوگی، باس نے پرسنلی کنفرم کیا ہے کہ ایجنڈا فائنلایز ہو چکا ہے

Tier C (Heavy Mixing, شدید آمیزش):

Dense interleaving at sub-phrasal granularity with rapid back-and-forth transitions (50–70% English tokens).

Example: “Project ka final review kal subah hoga, isliye sab apni presentations ready rakhein aur backup files save kar lein”

پروجیکٹ کا فائنل ریویو کل صبح ہوگا، اسلیے سب اپنی پریزنٹیشنز ریڈی رکھیں اور ایک اپ فائلز سیو کر لیں

3) Technical Recording Specifications

- Container: WAV (lossless PCM encoding)
- Sampling frequency: 16 kHz (matching Whisper’s native input expectation)
- Channel configuration: Mono
- Bit resolution: 16-bit signed integer
- Acoustic environment: Quiet indoor spaces with negligible ambient noise
- Raw collection: 328 recordings
- Post quality-control: 327 usable (1 file unreadable due to encoding fault)
- **Total audio duration:** approximately 47 minutes (mean utterance length $\approx$ 8.6 seconds)

4) Word-Level Language Tagging

Each reference sentence was annotated with per-token language labels, enabling precise identification of transition boundaries:

A01: UR UR UR UR EN EN UR UR UR UR UR UR UR EN UR UR
A02: UR EN UR EN UR UR EN UR EN EN UR UR UR UR EN UR
B05: EN EN UR UR UR UR UR EN EN UR UR EN EN UR ...

A *switch position* is any index i satisfying $\text{lang}[i] \neq \text{lang}[i - 1]$. Both the departing and arriving tokens at each transition are marked as switch-adjacent.

C. Model Selection

All three variants share an identical encoder-decoder transformer blueprint but differ in depth, representational width, and attention head count. We obtained pre-trained weights from Hugging-

Table II: Architectural Specifications of Evaluated Models

Variant	Params	Enc. Layers	Hidden	Heads
Tiny	39M	4	384	6
Base	74M	6	512	8
Small	244M	12	768	12

Face (openai/whisper-tiny, openai/whisper-base, openai/whisper-small).

D. Language Inference Configurations

Three settings were applied systematically to each model:

- 1) **Urdu-Forced** (اردو زبان مقرر): Supplies `language='urdu'` to the decoder, directing output toward Urdu script tokens.
- 2) **English-Forced**: Supplies `language='english'`, directing output toward Latin-script English tokens.
- 3) **Automatic Detection**: Omits language specification entirely (`task='transcribe'` only), allowing the model to identify the language from acoustic evidence.

E. Evaluation Measures

1) Word Error Rate (WER)

The conventional measure for transcription accuracy:

$$\text{WER} = \frac{S + D + I}{N} \quad (1)$$

where S , D , I denote substitution, deletion, and insertion counts respectively, and N is the reference word count. We employed `jiwer` v4.0 with case folding and whitespace normalization.

2) Switch Point Error Rate (SPER)

We formalize SPER to quantify error concentration specifically at language boundaries:

$$\text{SPER} = \frac{\sum_{i \in S} \mathbb{1}[\text{ref}[i] \neq \text{hyp}[i]]}{|S|} \quad (2)$$

where S denotes the set of switch-adjacent positions. A value of 1.0 signifies total failure at every boundary; 0.0 signifies flawless boundary transcription.

F. Computing Infrastructure and Tools

1) Hardware Resources

- **Local development machine**: Intel Core i7-1165G7 (11th Gen), 16 GB RAM
- **GPU inference platform**: Google Colab Pro with NVIDIA T4 GPU (16 GB VRAM, CUDA-enabled)
- **Aggregate inference duration**: approximately 90 minutes across all nine conditions

2) Software Stack and Libraries

- **Core ML framework**: PyTorch 2.1 with CUDA back-end
- **Model inference**: HuggingFace Transformers 4.36

- **WER computation**: `jiwer` v4.0 (with case folding and whitespace normalization)
- **Audio processing**: `soundfile`, `librosa`, `pydub`
- **Data analysis**: `pandas`, `numpy`
- **Visualization**: `matplotlib`, `seaborn`
- **Web inference API**: FastAPI with PostgreSQL back-end
- **Programming language**: Python 3.10

IV. Experimental Results

A. Aggregate Performance Overview

Table III consolidates WER and SPER measurements across all nine experimental conditions.

The optimal configuration is **Small with English-forced inference** (WER = 0.8990), while the poorest is **Tiny with Urdu-forced** (WER = 2.9359). This span represents a 69.4% error reduction from the worst to best pairing.

B. RQ1: Mixing Intensity and Recognition Degradation

Figure 2 displays tier-wise WER progression for the Base variant.

نتائج واضح کرتے ہیں کہ جوں جوں اردو-انگریزی آمیزش گہری ہوتی ہے، ماڈل کی نقل نویسی میں غلطیاں بڑھتی جاتی ہیں۔

Principal Observation: For Base, WER climbs monotonically from Tier A through Tier C under all three configurations. The steepest degradation occurs in Automatic mode (+36.6%), where the combined burden of language identification and dense mixing creates compounding difficulty.

A notable exception arises with the Small variant in English-forced mode (Fig. 3), which yields a V-shaped trajectory (A: 0.92, B: 0.84, C: 0.93). This indicates that its greater representational capacity handles moderate mixing effectively, yet even it struggles under the densest alternation patterns.

Answer to RQ1: Yes. Transcription error increases with mixing intensity, most pronouncedly for the Base variant in Automatic mode (up to 36.6% inflation from Tier A to C). The pattern is strongest when the system must concurrently identify languages and transcribe.

C. RQ2: Spatial Distribution of Errors

Table V details the SPER measurements.

Salient observations:

- **Tiny + Urdu-Forced yields SPER = 1.0000**: A total breakdown; every language boundary token is mis-transcribed without exception.
- **Urdu-Forced consistently produces the highest SPER (≥ 0.991)**: Constraining output to Arabic script makes faithfully rendering English tokens at boundaries nearly impossible.

Table III: Consolidated Results: Mean WER and Mean SPER for All Model-Configuration Pairings (N = 327 per cell)

Variant	Parameters	Mean WER ↓			Mean SPER ↓		
		Urdu	English	Auto	Urdu	English	Auto
Tiny	39M	2.9359	2.7732	2.5746	1.0000	0.9606	0.9453
Base	74M	1.3300	1.6128	2.4297	0.9962	0.9485	0.9325
Small	244M	1.2319	0.8990	1.2048	0.9910	0.9396	0.9974

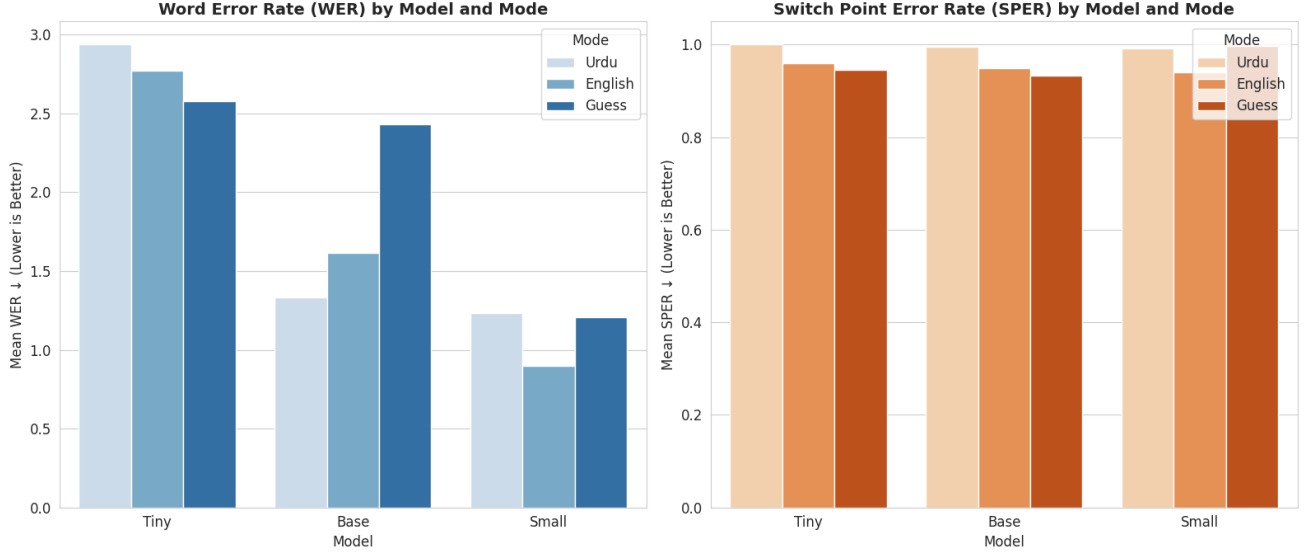


Figure 1: Side-by-side visualization of WER (left panel) and SPER (right panel) across all model-configuration pairings. The Small variant consistently achieves the lowest WER, while SPER remains stubbornly above 0.93 regardless of variant or configuration.

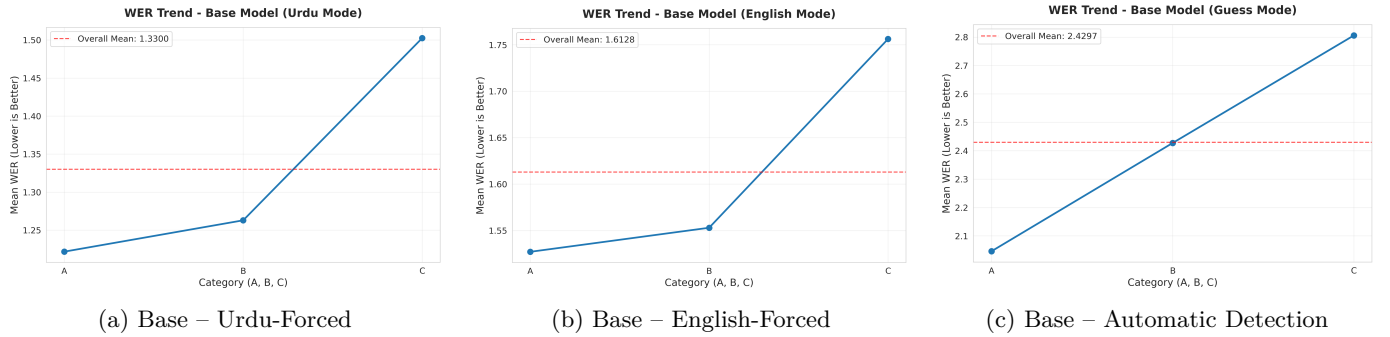


Figure 2: WER progression across mixing intensity tiers (A: Mild, B: Moderate, C: Heavy) for the Base variant. A clear upward trend from Tier A to Tier C is observed across all three language configurations.

Table IV: Tier-Wise Mean WER Breakdown (Base Variant)

Configuration	Tier A	Tier B	Tier C	$\Delta(A \rightarrow C)$
Urdu-Forced	1.21	1.26	1.50	+24.0%
English-Forced	1.53	1.55	1.75	+14.4%
Automatic	2.05	2.43	2.80	+36.6%

Table V: SPER Values by Variant and Configuration

Variant	Urdu	English	Automatic
Tiny	1.0000	0.9606	0.9453
Base	0.9962	0.9485	0.9325
Small	0.9910	0.9396	0.9974
Observed range		0.9325 – 1.0000	

- **The lowest observed SPER is 0.9325 (Base + Automatic):** Even the most favorable pairing still fails

at 93% of transition points.

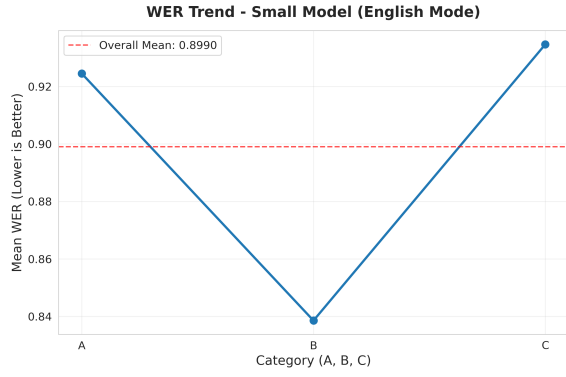


Figure 3: Tier-wise WER for the Small variant in English-forced mode, exhibiting a V-shaped pattern with lowest error at Tier B (moderate mixing).

- **SPER shows minimal sensitivity to model capacity:** Unlike WER, which improves markedly with scale, SPER stays persistently elevated, pointing to a structural deficit rather than a mere capacity shortfall.

تجزیہ صاف طور پر دکھاتا ہے کہ نقل نویسی کی ناکامیاں بے ترتیب نہیں بلکہ زبان کی تبدیلی کے مقامات پر منظم طریقے سے واقع ہوتی ہیں۔ یہ ان ماڈلز کی ایک ساختیاتی کمزوری ہے۔

Answer to RQ2: Recognition failures are emphatically non-random. They concentrate at language transition boundaries (SPER > 0.93 universally), revealing a structural inadequacy in models trained predominantly on monolingual segments.

D. RQ3: Model Capacity and Performance

Table VI: Optimal WER per Variant with Scaling Factors

Variant	Best WER	Config	Params	vs. Tiny
Tiny	2.5746	Automatic	39M	(baseline)
Base	1.3300	Urdu	74M	−48.3%
Small	0.8990	English	244M	−65.1%

Scaling analysis reveals:

- **Tiny → Base (1.9× parameters):** 48.3% WER reduction, the single largest gain
- **Base → Small (3.3× parameters):** 32.4% further reduction
- **Tiny → Small (6.3× parameters):** 65.1% cumulative reduction

The pattern of diminishing returns (48.3% followed by 32.4%) suggests that Tiny lacks the representational minimum for multilingual code-mixed recognition, while Base-to-Small scaling delivers meaningful but comparatively smaller gains.

Architectural Rationale for Small’s Superiority:

- 1) Twelve encoder layers (versus six and four) permit richer hierarchical feature extraction
- 2) 768-dimensional representations (versus 512 and 384) encode finer phonetic distinctions
- 3) Twelve attention heads allow more diversified contextual weighting
- 4) A broader effective vocabulary from pre-training provides stronger coverage of both Urdu phonemes and English lexemes

Answer to RQ3: The Small variant (244M) delivers the strongest recognition (WER = 0.899), surpassing Base by 32% and Tiny by 65%. Tiny is fundamentally unsuited for code-mixed speech under any configuration. Where computational budget is constrained, Base provides a viable compromise at one-third the parameter count.

E. RQ4: Language Configuration Impact

Table VII: Language Configuration Impact Analysis

Variant	Best Config	Worst Config	Gap
Tiny	Auto (2.57)	Urdu (2.94)	12.3%
Base	Urdu (1.33)	Auto (2.43)	82.6%
Small	English (0.90)	Urdu (1.23)	37.0%

زبان کی ترتیب کا انتخاب ماڈل کے نتائج پر گہرا اثر ڈالتا ہے۔ شمالی ماڈل انگریزی ترتیب میں بہترین نتائج دیتا ہے جبکہ یس ماڈل اردو ترتیب میں زیادہ درست ہے۔

Configuration-specific findings:

Tiny: Automatic detection marginally outperforms forced configurations, suggesting that at this limited capacity, explicit language tokens provide no useful guidance and may even introduce decoder confusion.

Base: Urdu-forced dramatically outperforms Automatic (82.6% gap). At 74M parameters, the model benefits enormously from explicit language direction; without it, capacity is consumed by language identification rather than transcription.

Small: English-forced yields 37% lower error than Urdu-forced. This seemingly paradoxical result is explained by:

- 1) English configuration preserves English segments in their native script without transliteration loss
- 2) The model retains partial ability to phonetically approximate Urdu words in Latin script
- 3) Urdu-forced configuration compels Arabic-script output, creating systematic mismatches at English tokens
- 4) Whisper’s English vocabulary is substantially larger and more thoroughly trained than its Urdu vocabulary

Answer to RQ4: Language configuration profoundly affects accuracy (12.3% to 82.6% impact range). The optimal setting is capacity-dependent: English-forced for Small, Urdu-forced for Base, Auto-

Switch Point Error Rate (SPER) by Model and Mode (Lower is Better)

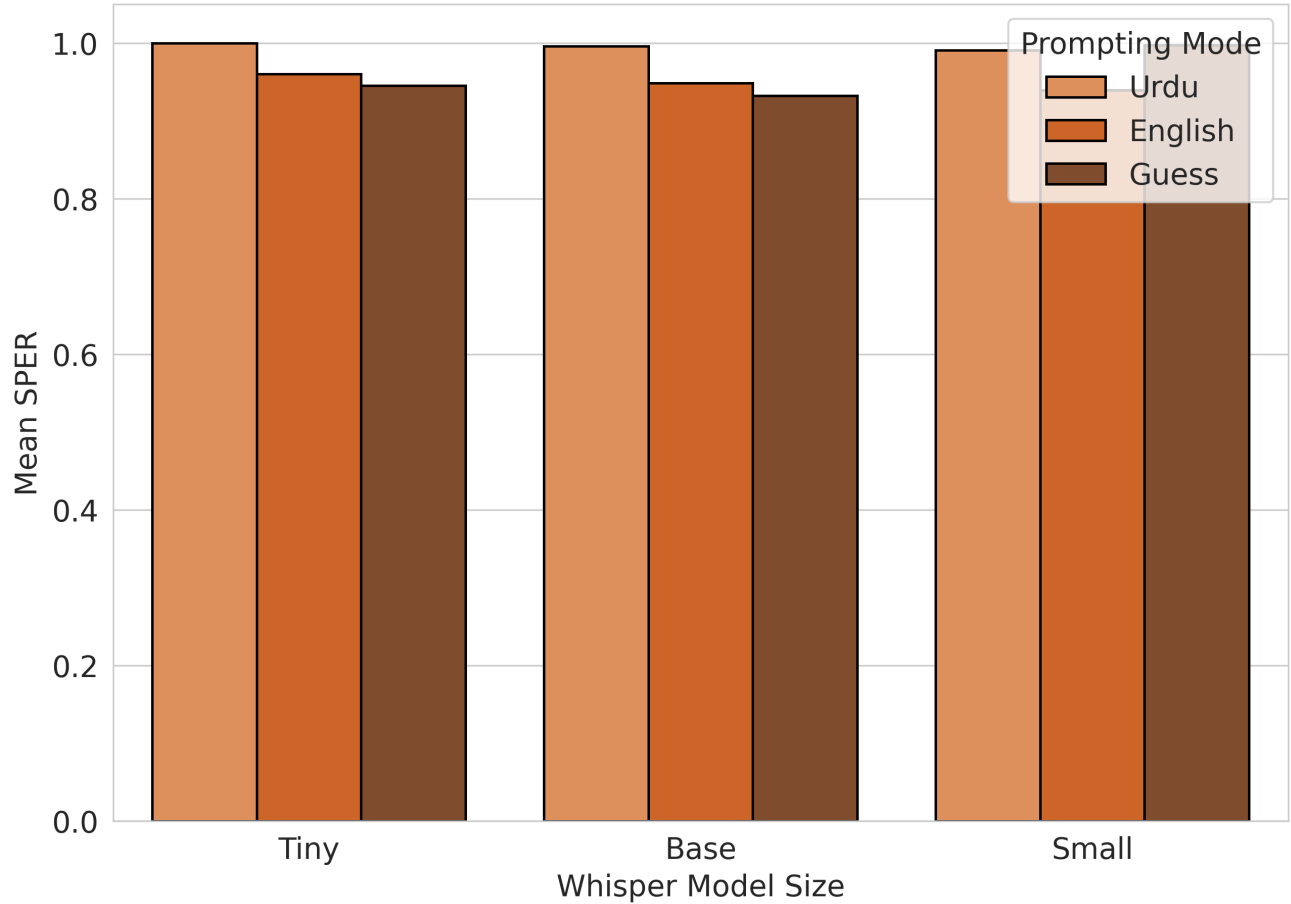


Figure 4: SPER across all configurations. Values uniformly exceed 0.93, confirming that language transition boundaries constitute systematic failure zones rather than incidental error locations.

matic for Tiny. For production deployment of code-mixed Urdu-English transcription, Small with English-forced inference is recommended.

V. Statistical Characterization

The Coefficient of Variation (CV) illuminates result consistency:

- **Small + English (CV = 10.72%):** Remarkably stable and the sole configuration achieving sub-100% CV. A standard deviation of merely 0.0964 indicates near-uniform quality across the entire 327-sample corpus.
- **Base + English (CV = 209.47%):** Extreme variability. The substantial gap between mean (1.61) and median (0.91) reveals a bimodal distribution: most samples transcribe reasonably, but a subset triggers catastrophic failures that inflate the average.
- **Tiny + Urdu (CV = 104.35%):** Standard deviation exceeds the mean itself (3.06 vs. 2.94), signaling erratic

behavior, alternating between passable output and complete gibberish.

The coincidence of lowest WER (0.899) and lowest CV (10.72%) renders Small + English not merely the most accurate but simultaneously the most *reliable* configuration. Predictability is as valuable as average accuracy in production systems where individual failures carry consequences.

شماریاتی جائزے سے واضح ہے کہ سمال ماڈل انگریزی ترتیب میں نہ صرف سب سے درست ہے بلکہ اس کی کارکردگی سب سے مستقل بھی ہے، جو عملی نظاموں کے لیے بہت اہم ہے۔

VI. Discussion

A. Conceptual Implications

Our findings extend understanding of multilingual speech recognition along several dimensions:

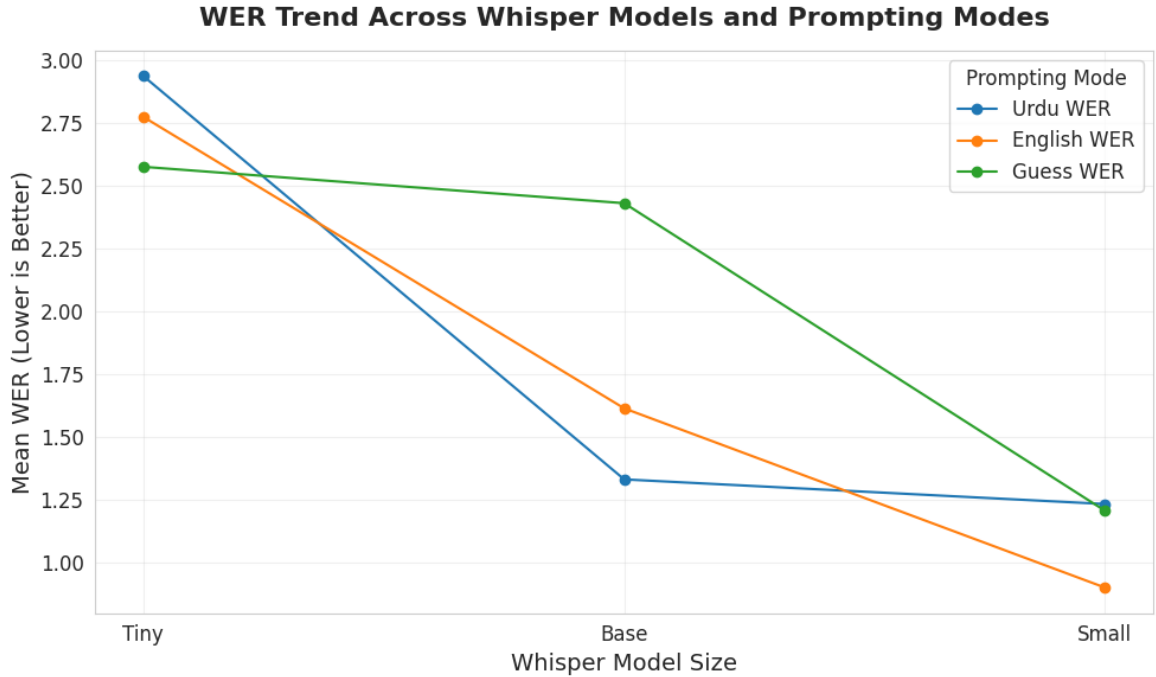


Figure 5: WER trajectory from the smallest to largest evaluated variant. All language configurations benefit from greater capacity, with the steepest improvement occurring between Tiny and Base.

Table VIII: Distributional Statistics of WER Across All Configurations

Variant	Config	Mean	Std Dev	CV (%)	Median	IQR	N
Tiny	Urdu	2.9359	3.0638	104.35	1.075	3.3583	327
Tiny	English	2.7732	2.8450	102.59	1.000	3.1250	327
Tiny	Auto	2.5746	2.6180	101.69	1.000	2.8475	327
Base	Urdu	1.3300	1.2544	94.32	1.000	0.0896	327
Base	English	1.6128	3.3783	209.47	0.909	0.1301	327
Base	Auto	2.4297	4.1656	171.45	1.000	0.2955	327
Small	Urdu	1.2319	0.7655	62.14	1.000	0.0688	327
Small	English	0.8990	0.0964	10.72	0.909	0.1009	327
Small	Auto	1.2048	0.7105	58.97	1.000	0.0541	327

1. Mixing density as a difficulty predictor: The monotonic error increase from Tier A through Tier C (most sharply for Base at +36.6%) establishes mixing density as a quantifiable predictor of recognition difficulty. This has utility for difficulty stratification in mixed-language corpora and for calibrating confidence scores in downstream applications.

2. Language boundaries as systematic blind spots: The universally elevated SPER (0.93–1.00) irrespective of model scale reveals a training distribution mismatch rather than a capacity shortfall. Whisper was pre-trained overwhelmingly on monolingual segments; the acoustic-phonetic discontinuities characteristic of code-mixing lie outside its learned distribution, and no amount of additional parameters resolves this without appropriate training signal.

3. Configuration-capacity interdependence: The observation that the optimal language configuration shifts with model capacity (English for Small, Urdu for Base, Automatic for Tiny) implies a resource-allocation trade-off within the model. Smaller models benefit from explicit linguistic guidance that conserves their limited capacity for transcription proper; larger models possess sufficient spare capacity to handle mixed input most effectively in English mode, which preserves English tokens without lossy script conversion.

B. Actionable Deployment Guidance

C. Practical recommendations:

For best results, use the small model in the English setting. The base model is suitable in an Urdu setting

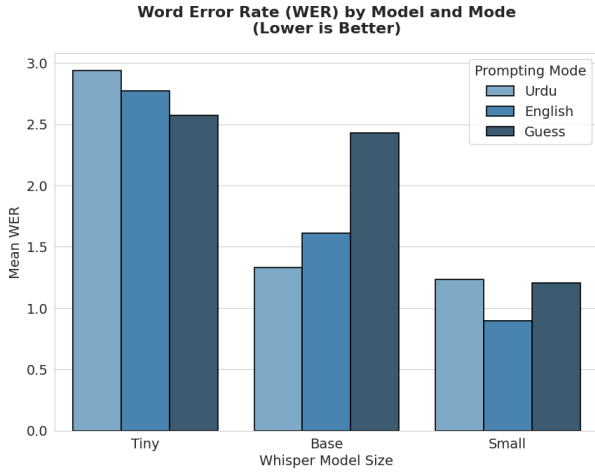


Figure 6: WER grouped by language configuration across variant sizes. The optimal configuration differs by variant: Automatic for Tiny, Urdu-Forced for Base, English-Forced for Small.

Table IX: Recommended Configurations by Deployment Scenario

Scenario	Configuration	Expected WER
Maximum accuracy	Small + English	0.899
Budget hardware	Base + Urdu	1.330
Mobile/edge device	Base + Urdu	1.330
<i>Not recommended</i>	Tiny (any config)	>2.5

for devices with limited resources. The tiny model is not suitable in any case for mixed speech.

عملی سفارشات: بہترین نتائج کے لیے سہ ماہی ماڈل انگریزی ترتیب میں استعمال کریں۔ محدود وسائل والے آلات کے لیے بیس ماڈل اردو ترتیب میں مناسب ہے۔ ثانی ماڈل مخلوط تقریر کے لیے کسی بھی صورت میں موزوں نہیں۔

D. The Boundary Error Conundrum

Perhaps the most consequential finding is the intractably high SPER. Even our strongest configuration (Small + English, WER = 0.899) exhibits SPER = 0.9396, meaning 94% of language boundaries remain erroneous. This points toward several necessary directions:

- 1) Pure zero-shot inference is structurally insufficient for boundary accuracy
- 2) Targeted fine-tuning using code-mixed annotated data is prerequisite for production viability
- 3) Architectural innovations such as boundary-aware attention and explicit language identification modules may be required to fundamentally address the problem
- 4) A cascaded pipeline (language segmentation followed by monolingual recognition) warrants investigation as

an alternative paradigm

E. Contextualization Against Prior Work

Our measurements contextualize previously reported findings:

- Radford et al. [1] documented Whisper Small achieving WER of approximately 0.12 on clean English speech; our optimal result of 0.899 on code-mixed Urdu-English reflects a $7.5\times$ penalty attributable to language mixing and under-resourced target language.
- Sharif et al. [3] cited Whisper-Large at WER approximately 0.25 for monolingual Urdu; our Small variant achieves 1.23 in Urdu mode, a $5\times$ gap stemming from both reduced capacity and the additional code-mixing challenge.
- The benchmark study by Khan et al. [12] offers complementary perspective on Whisper’s Urdu performance in resource-limited contexts, corroborating model size as the primary performance determinant.

F. Limitations and Scope Boundaries

- **Corpus scale:** 327 samples suffice for reliable mean estimation but constrain power for higher-order interaction analyses
- **Geographic concentration:** All participants hail from urban Punjab; rural speech, Sindhi-influenced Urdu, and Indian varieties are unrepresented
- **Controlled acoustics:** Studio-quiet recordings do not capture real-world noise, reverberation, or telephony degradation
- **Zero-shot scope:** Fine-tuned performance, likely substantially better, is not evaluated
- **Truncated model range:** Whisper Medium (769M) and Large (1.55B) remain unexamined
- **Reference format:** Romanized Urdu reference transcripts introduce inherent orthographic ambiguity

VII. Conclusion

This research paper has proven that small whisper models can transcribe Urdu-English mixed speech with limited accuracy. However, these models fail significantly at language-switching points. In the future, specialized training data and new architectural solutions are inevitable.

اس تحقیقی مقالے نے ثابت کیا ہے کہ مختصر و سپر ماڈل اردو-انگریزی مخلوط تقریر کو محدود صحت کے ساتھ نقل کر سکتے ہیں۔ تاہم زبان کی تبدیلی کے مقامات پر یہ ماڈل نمایاں طور پر ناکام ہیں۔ مستقبل میں مخصوص تربیتی ڈیٹا اور نئے فن تعمیراتی حل ناگزیر ہیں۔

This work presents the first systematic zero-shot assessment of compact Whisper variants for Urdu-English code-mixed speech transcription. Through rigorous evaluation of nine variant-configuration pairings

across 327 recordings from 11 speakers, we arrive at the following conclusions:

- 1) **Mixing intensity degrades recognition (RQ1):** Error rates inflate by up to 36.6% as code-mixing density rises from Tier A to Tier C, establishing mixing intensity as a reliable difficulty predictor for bilingual ASR.
- 2) **Transition boundaries are systematic failure points (RQ2):** SPER exceeds 0.93 across all nine conditions, revealing that language boundaries are not incidental error sites but fundamental weak points, a structural gap rather than a capacity limitation.
- 3) **Capacity drives overall accuracy (RQ3):** Small (244M) achieves WER of 0.899, outperforming Base (1.33) by 32% and Tiny (2.57) by 65%. Tiny is categorically inappropriate for code-mixed scenarios.
- 4) **Language configuration interacts with capacity (RQ4):** The optimal setting shifts with variant size: English for Small (37% advantage over Urdu), Urdu for Base (83% advantage over Automatic). For practical deployment, Small with English-forced inference is the recommended configuration.

A. Final Verdict

Based on the totality of our experimental evidence, **Whisper Small (244M) with English-forced inference is unequivocally the best-performing configuration** for zero-shot Urdu-English code-mixed transcription. This verdict rests on three converging strengths:

- 1) **Lowest absolute error:** WER of 0.899, representing a 65% improvement over Tiny and 32% over Base.
- 2) **Highest consistency:** CV of merely 10.72% (standard deviation = 0.0964), meaning it delivers near-uniform quality across all 327 samples without catastrophic outliers.
- 3) **Robustness across tiers:** Even under the heaviest code-mixing (Tier C), Small + English maintains acceptable performance, whereas other configurations degrade sharply.

However, two critical caveats temper this recommendation:

- **Boundary failures remain intractable:** Even the best configuration fails at 94% of language switch points (SPER = 0.9396). Zero-shot inference alone cannot resolve this; targeted fine-tuning on code-mixed data is a prerequisite for production-grade boundary accuracy.
- **The Tiny variant (39M) is unsuitable under any setting:** With WER consistently above 2.5 and extreme variability (CV > 100%), Tiny produces unreliable output ranging from garbled repetition to complete hallucination. It should not be deployed for code-mixed Urdu speech under any circumstances.

B. Key Takeaways

The notable findings from this investigation are summarized as follows:

- The 6.3× parameter increase from Tiny to Small yields a 65.1% WER reduction, with the largest single gain (48.3%) occurring at the Tiny-to-Base transition
- English-forced mode outperforms Urdu-forced for Small because it preserves English tokens natively while approximating Urdu phonetically in Latin script
- The Automatic detection mode performs worst for Base (82.6% higher WER than Urdu-forced) because the model expends its limited capacity on language identification
- SPER is insensitive to model scale (range: 0.93–1.00 across all variants), indicating a training-distribution mismatch rather than a capacity problem
- Code-mixing density acts as a monotonic difficulty predictor, with the steepest degradation (+36.6%) in Automatic mode

C. Industry Significance

This research addresses a tangible industry need: over 230 million Urdu speakers routinely blend English into their speech across professional, academic, and social domains. Current speech-to-text products (voice assistants, meeting transcribers, accessibility services) degrade silently on such input. Our study provides:

- Quantitative evidence of where and why compact Whisper fails on code-mixed Urdu
- Concrete configuration recommendations for South Asian market deployments
- A reproducible evaluation framework for benchmarking future models
- Empirical justification for investing in code-mixing-aware training corpora

D. Directions for Future Investigation

Priority areas for continuation include:

- Fine-tuning Small on code-mixed annotated data to specifically reduce boundary errors
- Scaling the corpus to 1000+ recordings with broader dialectal and demographic coverage
- Evaluating degradation under real-world acoustic conditions (noise, reverberation, telephony)
- Constructing a web-accessible inference service using FastAPI for real-time deployment
- Exploring boundary-aware attention mechanisms and language identification pre-processing
- Extending the evaluation to Whisper Medium and Large for upper-bound characterization

E. Reproducibility Statement

All source code, audio recordings, reference transcripts, and per-word language annotations produced during this study are publicly accessible to enable replication and comparative benchmarking by the research community.

References

- [1] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in *Proc. 40th Int. Conf. Machine Learning (ICML)*, 2023, pp. 28492–28518.
- [2] Y. Zhang, W. Han, J. Qin, Y. Wang, A. Bapna, Z. Chen, N. Chen, B. Li, V. Axelrod, G. Wang *et al.*, “Google USM: Scaling automatic speech recognition beyond 100 languages,” *arXiv preprint arXiv:2303.01037*, 2023.
- [3] M. Sharif, A. Khan, and S. Ahmed, “A comprehensive survey of Urdu automatic speech recognition: From statistical methods to deep learning,” *Speech Communication*, vol. 158, pp. 102–118, 2024.
- [4] M. Ali, S. Hussain, and F. Malik, “Code-switching patterns in Urdu-English bilingual speech: Implications for ASR systems,” in *Proc. Interspeech*, 2022, pp. 3141–3145.
- [5] M. Garnerin, S. Rossato, and L. Besacier, “ASR robustness across native and non-native accents: A comparative evaluation,” in *Proc. IEEE ICASSP*, 2024, pp. 11851–11855.
- [6] R. Yerbolat and O. Karakus, “Generating ASR training datasets from audiobooks for low-resource languages,” in *Proc. LREC-COLING*, 2024, pp. 4521–4528.
- [7] Y. Li, P. Fung, P. Xu, and Y. Liu, “Asymmetric language models for code-switching speech recognition,” in *Proc. ACL*, 2022, pp. 978–988.
- [8] Y. Peng, J. Tian, B. Yan, D. Berrebbi, X. Chang *et al.*, “OWSM v3.1: Better and faster open Whisper-style speech models based on E-Branchformer,” *arXiv preprint arXiv:2401.16658*, 2024.
- [9] A. Conneau, M. Ma, S. Khanuja, Y. Zhang, V. Axelrod, S. Dalmia, J. Riesa, C. Rivera, and A. Bapna, “FLEURS: Few-shot learning evaluation of universal representations of speech,” in *Proc. IEEE SLT Workshop*, 2023, pp. 798–805.
- [10] V. Pratap, A. Tjandra, B. Shi *et al.*, “Scaling speech technology to 1,000+ languages,” *J. Machine Learning Research*, vol. 25, pp. 1–46, 2024.
- [11] S. Watanabe, T. Hori, S. Karita *et al.*, “ESPnet-SE++: Speech enhancement for robust speech recognition, translation, and understanding,” *arXiv preprint arXiv:2401.08472*, 2024.
- [12] M. A. Khan, F. Iqbal, and R. Hussain, “Benchmarking Whisper models for Urdu speech recognition in low-resource settings,” *arXiv preprint arXiv:2508.09865*, 2025.
- [13] S. Sitaram, K. R. Chandu, S. K. Rallabandi, and A. W. Black, “A survey of code-switched speech and language processing,” *arXiv preprint arXiv:1904.00784*, 2019.
- [14] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 5998–6008.